

Native TERR ON ND and the A380X VD terrain profile

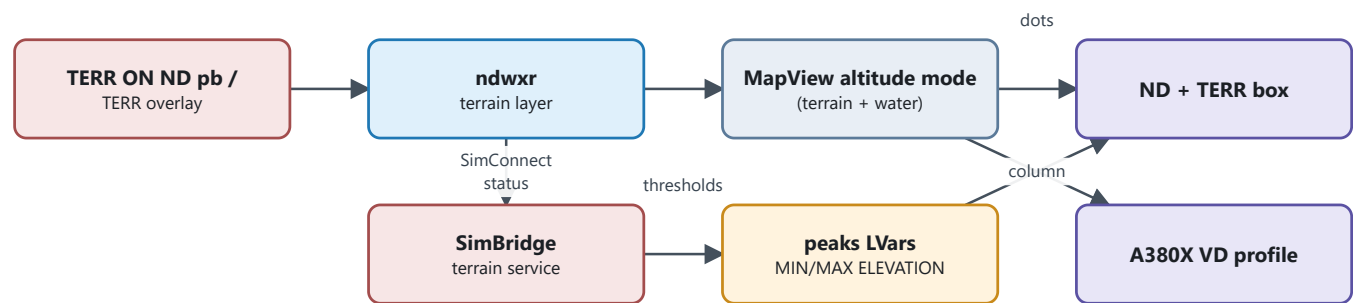
EGPWS-style terrain from MSFS terrain data, without SimBridge rendering

Branch	feature/nd-terrain
Aircraft	A32NX and A380X
Commits	tip acffe345b (PR 4)
Base	feature/a380x-nd-weather-radar
Status	In master. Flown. PR draft 04.

Specification

- An altitude-mode MapView colours the terrain by height relative to the aircraft in 250 ft bands (-29 500 to +2 500 ft, 128 entries); the EGPWS look is built from the bands: red from 2000 ft above, yellow from 1000 ft above to 500 ft below (250 ft gear down), green to 1000 / 2000 ft below; random stipple dots from blend-only passes.
- Reference altitude 30 s ahead when descending faster than 1000 ft/min (A320 FCOM DSC-31-45).
- Water from a second altitude view, drawn as teal dots. Map rotated by the ND true heading; every map page but PLAN, also on the ground, in place of the weather.
- A32NX gated by the TERR ON ND pb; A380X by the TERR overlay with a WXR & TAWS system selected.
- A380X VD terrain profile: a third gauge per ND side, the terrain column along the cut stretched over the plot and filled brown.
- Peaks box: figures from SimBridge over SimConnect (ndwxr replaces the stock terronnd as SimBridge's client); terronnd no longer instantiated.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

MSFS limits found in-sim: at most 8 MapViews per module (a ninth crashes the draw), no growing the module memory while drawing (16 MB reserved).

PR description (from the PR draft)

From `pr-drafts/04-terrain-on-nd.md`, section "Summary of Changes" onwards.

Summary of Changes

Terrain on the ND (TERR ON ND) without SimBridge, drawn by the `ndwxr` module for both aircraft from MSFS's own terrain data, plus a terrain profile on the A380X VD.

How the ND terrain works

- An altitude-mode MapView colours the terrain by the aircraft's altitude minus the terrain height in 250 ft bands, over -29,500 to +2,500 ft (128 entries). The EGPWS look is built from the bands: dense red from 2000 ft above the aircraft, dense/medium yellow from 1000 ft above, medium yellow down to 500 ft below (250 ft with the gear down), dense green to 1000 ft below (500 ft), light green to 2000 ft below and nothing under that. The dots are a random stipple (~1 px cells, ~70% lit when dense) made with blend-only passes (ordered compare against a noise image, sharpened by repeated squaring), as MSFS gives a gauge no way to read the texture back.
- Water never takes the altitude list's colours (it always gets the first entry, like terrain far above), so a second altitude view with a two-entry list tells water from land; the water is wiped from the terrain and redrawn as cyan-blue dots at every altitude.
- The map is rotated by the ND's own true heading. ARC shows the forward half disk, the ROSE modes the whole disk; every map page but PLAN, also on the ground, in place of the weather.
- A32NX: gated by `A32NX_EGPWC_ND_{L,R}_TERRAIN_ACTIVE` (the TERR ON ND pb), gear from `A32NX_EGPWC_GEAR_IS_DOWN`. A380X: gated by the ND overlay being TERR with a WXR & TAWS system selected and not failed.
- Terrain far above the range's top does not clamp to the first entry in MSFS (it comes out as rings of the other entries), which is why the range reaches -29,500 ft and every band above +2000 ft is red: nothing on Earth is higher above the aircraft.

A380X VD terrain profile: a third `ndwxr` gauge per ND (`LV / RV` in `panel.cfg`, its own surface). The heading-line column of a terrain view whose altitude range follows the VD's limits is stretched over the plot height, compared with a vertical ramp and filled brown; water is cyan from sea level down. Needs the TERR SYS button of the SURV page not to be OFF.

MSFS limits that shaped the module (found by crashing the gauge draw in-sim): a module can have at most 8 MapViews (a ninth is created without complaint and crashes the draw; hiding views does not help). The A32NX uses 4 per ND; the A380X has 2 per ND, which are either the weather pair or the terrain pair (never wanted together) and are reconfigured when the crew switches, and 2 per VD terrain gauge. Growing the module's memory while it draws also crashes it, so `build.sh` reserves 16 MB up front.

Peaks box (PR 9's commit 43a6899f1, 2026-09-23): the TERR box with the highest and lowest elevation of the range needs real elevations, which a gauge cannot read back from MSFS, so its figures come from SimBridge, over SimConnect: SimBridge's terrain service only renders while a SimConnect client takes part (its HTTP `renderingThresholds` endpoint stays empty while it is connected to the sim; an HTTP poller was tried first and reverted), and the stock `terronnd` gauge was that client. `ndwxr` now takes its place from the first ND gauge installed: it creates and writes the aircraft status block `FBW_SIMBRIDGE_EGPWC_AIRCRAFT_STATUS` every 100 ms (the content of `terronnd's collection.cpp`, with the terrain reported on per side exactly when the module selects it), subscribes to the two threshold areas SimBridge writes per ND and publishes the figures on the four `L:A32NX_EGPWC_ND_{side}_TERRAIN_{MIN,MAX}_ELEVATION(_MODE)` LVars that `TerrainMapThresholds` already shows, while terrain is selected on that side on a map page and the figures are fresh (5 s) and for the ND's range and mode; -1 otherwise. It reconnects every 5 s and disconnects when the owning gauge is killed. `build.sh` adds the SimConnect SDK include path; the client data layouts are `terronnd's simbridge.h`. Without SimBridge the box stays empty and the terrain is drawn as before. (Peaks mode, the display when all terrain is far below the aircraft, uses the same figures: PR 9.)

terronnd retired (uncommitted, 2026-09-23): the stock `terronnd` gauge is removed from both `panel1.cfg` (the `was` is still built, nothing instantiates it). Reasons found in test with SimBridge 0.7.0 on MSFS 2024: its picture is not clipped at the compass arc (the old `ARC_CLIP` dome), it is never expired (the last frame stays after SimBridge closes), and on the A380X its SimConnect status reports terrain on for an ND whenever the VD wants it, which SimBridge falls back to after two minutes without HTTP posts (a stationary aircraft), so the F/O terrain stayed on with the overlay off. The native picture is the only one now; SimBridge only supplies the peaks figures. The A32NX gets `TawsStatusBridge` (systems host) to post the aircraft status over HTTP as the A380X does (`terronnd` used to send it over SimConnect; the SimConnect status block SimBridge takes the aircraft's position from is written by `ndwxr` again, see the peaks box), and both hosts re-post it every 30 s so SimBridge never drops the HTTP source. The ARC picture (terrain and weather) leaves the ND's bottom-right corner free (same corner as CanvasMap's `ARC_CLIP`) for the TERR box and the radar mode text.

Not done: the VD profile follows the heading line and not the flight plan path, and switching between weather and terrain on the A380X ND takes about a second.

Cockpit API Changes

No new LVars. The terrain gauge reads `L:A32NX_EGPWC_ND_{L,R}_TERRAIN_ACTIVE`, `L:A32NX_EGPWC_GEAR_IS_DOWN`, `L:A32NX_GPWS_TERR_OFF`, `L:A32NX_TERR_{1,2}_FAILED`, `L:A32NX_WXR_TAWS_SYS_SELECTED`, `L:A380X_EFIS_{L,R}_ACTIVE_OVERLAY` and `L:A32NX_VD_{1,2}_RANGE_LOWER/UPPER` (the last two pairs already exist), `panel1.cfg` gets the VD gauges.

Screenshots (if necessary)

TO ADD: TERR ON ND on both aircraft in ARC and ROSE, at the same place before/after (before: no terrain without SimBridge), e.g. KASE (Aspen, 7,820 ft) at 40 and 320 NM, and LSZC; sea and lakes in blue; gear up/down colour change; the A380X VD terrain profile; weather on one ND and terrain on the other.

References

- MSFS SDK, `MSFS_MapView.h` altitude view mode (implementation).
- TO ADD:** the Honeywell EGPWS / A320 & A380 FCOM references for the terrain colours and the altitude thresholds (the values above), and for the gear-down thresholds; if the blue water cannot be referenced, drop it (it was added because the reference aircraft in the sim show it).

Additional context

The stock `terronnd` module is only a client of the external SimBridge program; its source is not changed by this PR, but the gauge is no longer instantiated (removed from both `panel1.cfg`, see above), so there is one terrain layer. Say in the review if the `terronnd` build task should go too. After the range fix the A380X VD terrain profile still uses a range that ends at the top of its plot (same weakness as the ND had), it was not affected in tests at Aspen where the plot sits above the terrain.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

- A32NX: TERR ON ND pb ON, ND in ARC or ROSE NAV: terrain in red/yellow/green dots around the aircraft, rotating with the heading. Check at high terrain: on the ground at KASE (Aspen) at 40 and 320 NM there must be red.
- Lower the gear: the medium-yellow/green limits move (250/500 ft instead of 500/1000 ft below). Water is blue dots.
- PLAN: no terrain. On the ground the terrain is drawn.
- A380X: TERR overlay on the EFIS panel and a WXR & TAWS system selected: same on the ND; VD shows the terrain profile (TERR SYS not OFF); with TERR SYS OFF the VD says NO TERR DATA AVAILABLE (see the SURV PR).
- A380X: WX on the CPT ND and TERR on the F/O ND at once, then swap them (about 1 s to switch).
- Weather and terrain never both show on the same ND; the weather comes back when TERR is turned off.